MAHATMA GANDHI UNIVERSITY, KOTTAYAM

UGP (HONOURS) REGULAR EXAMINATION – MARCH 2026

Fourth Semester B.Sc. Degree Examination

Course: Applied Cryptography

Course Code: MG4DSCCFS201

SCHEME

Part A

Short answer type questions

Answer any 10 questions.each carries 1 mark each

1. Define brute-force attack.

A brute-force attack is a method of breaking encryption by trying all possible keys until the correct key is found.

2. Write the number of rounds in AES-128, AES-192, and AES-256.

- AES-128 → 10 rounds
- AES-192 → 12 rounds
- AES-256 → 14 rounds

3. State the need for digital certificates.

Digital certificates are used to verify the identity of users, servers, or organizations and to establish secure communication.

4. Define substitution cipher.

A substitution cipher is an encryption technique in which each plaintext character is replaced by another character according to a fixed rule.

5. Define stream cipher.

A stream cipher encrypts data one bit or one byte at a time using a keystream.

6. Name the two main phases of sponge construction.

1. Absorbing Phase
2. Squeezing Phase

7. Discuss how cryptography ensures security in online banking.

Cryptography protects banking data through encryption, authentication, and digital signatures, ensuring confidentiality and integrity.

8. Identify the main security risk in symmetric encryption.

The main security risk is secure key distribution and management, since both parties use the same secret key.

9. List the public parameters required in the Diffie–Hellman algorithm.

- A large prime number (p)
- A generator (g)

10. Which cryptographic primitive is used to ensure data confidentiality?

Encryption is the cryptographic primitive used to ensure data confidentiality.

11. Define ECB.

Electronic Codebook (ECB) is a block cipher mode in which each plaintext block is encrypted independently using the same key.

12. Define a key exchange protocol.

A key exchange protocol is a method that allows two parties to securely establish a shared secret key over an insecure network.

Part B

Short Essay Questions (5 Marks)

Answer any 4 Questions

13. Explain the use of hash functions in digital signatures.

A hash function converts a message of any length into a fixed-length hash value (message digest). In digital signatures, the sender first computes the hash of the message and then encrypts the hash using their private key to create the digital signature.

Process:

1. Create a hash of the message.
2. Encrypt the hash using the sender's private key.
3. Send the message and signature.
4. Receiver decrypts the signature using the sender's public key.
5. Receiver computes the hash of the received message.
6. If both hashes match, the signature is valid.

Uses:

- Ensures data integrity.
- Verifies sender authenticity.
- Provides non-repudiation.
- Reduces computation time compared to signing the entire message.

14. Explain DES operation in detail.

DES (Data Encryption Standard) is a symmetric-key block cipher that encrypts 64-bit blocks using a 56-bit key.

Operation:

1. Apply Initial Permutation (IP) to the plaintext.
2. Divide the block into Left (L) and Right (R) halves of 32 bits each.
3. Perform 16 rounds of encryption:
    - Expand R from 32 bits to 48 bits.
    - XOR with the round key.
    - Pass through S-boxes.
    - Apply permutation.
    - XOR with L.

- Swap halves.

4. After 16 rounds, combine the halves.
5. Apply Final Permutation (FP) to obtain ciphertext.

Features:

- Symmetric encryption algorithm.
- Uses 16 Feistel rounds.
- Widely used before AES.

15. Write the steps of key generation in RSA Algorithm.

RSA Key Generation Steps:

Select two large prime numbers, p and q.

Compute:$n = p \times q$

Compute Euler's Totient:$\phi(n) = (p-1)(q-1)$

Choose an integer e such that:$1 < e < \phi(n)$

and gcd(e, φ(n)) = 1.

Compute d such that:$d \times e \equiv 1 (\mathrm{mod} \phi(n))$

Public Key = (e, n)

Private Key = (d, n)

- Public key is used for encryption.
- Private key is used for decryption.

16. Explain the concept of asymmetric key cryptography.

Asymmetric cryptography uses two different but mathematically related keys: a public key and a private key.

Working:

- The public key is shared openly.
- The private key is kept secret.
- Data encrypted with the public key can only be decrypted with the private key.

Advantages:

- Solves key distribution problems.
- Supports digital signatures.
- Provides authentication and non-repudiation.

Examples:

- RSA
- Diffie–Hellman
- ECC

Applications:

- Secure email
- SSL/TLS
- Digital certificates

17. Write the advantages of CBC over ECB.

CBC (Cipher Block Chaining) is more secure than ECB (Electronic Codebook).

Advantages:

1. Identical plaintext blocks produce different ciphertext blocks.
2. Hides data patterns effectively.

3. Provides better confidentiality.
4. Resistant to pattern analysis attacks.
5. Uses an Initialization Vector (IV) for added security.

18. Explain the steps involved in generating and verifying an HMAC.

HMAC (Hash-based Message Authentication Code) provides message authentication and integrity using a secret key and hash function.

HMAC Generation

1. Sender selects a secret key K.
2. Combine the key with the message using inner padding (ipad).
3. Apply the hash function.
4. Combine the result with outer padding (opad).
5. Hash again to generate the HMAC value.
6. Send the message along with the HMAC.

HMAC Verification

1. Receiver receives the message and HMAC.
2. Uses the same secret key to compute a new HMAC.
3. Compares the computed HMAC with the received HMAC.
4. If both match, the message is authentic and unaltered.

Advantages:

- Ensures integrity.
- Provides authentication.
- Resistant to forgery attacks.

PART C

Essay type questions

Answer any 2 questions

Each question carries 10 marks

19. Symmetric Key Cryptography

    1. Same key used for encryption and decryption.
    2. Sender and receiver share a secret key.
    3. Fast and efficient for large data.
    4. Main issue: key distribution.
    5. Examples: AES, DES, Blowfish.

20. Block Cipher Modes of Operation

    1. ECB – Encrypts each block independently.
    2. CBC – Uses previous ciphertext block and IV.
    3. CFB – Converts block cipher to stream cipher.
    4. OFB – Uses keystream; no error propagation.
    5. CTR – Uses counter values; supports parallel processing.

21. Public Key Infrastructure (PKI)

    1. Framework for managing public keys and certificates.
    2. Provides authentication and secure communication.
    3. Components: CA, RA, Digital Certificates.
    4. Supports integrity and non-repudiation.
    5. Applications: SSL/TLS, digital signatures, secure email.